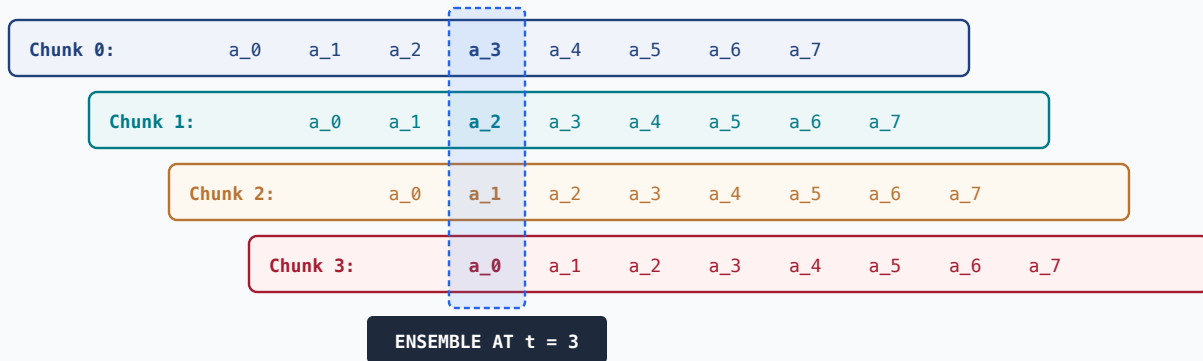


# RECEDING-HORIZON ACTION CHUNKING WITH EXPONENTIAL TEMPORAL ENSEMBLING

Blending overlapping trajectory proposals with exponential weight decay  $w_i = \exp(-\lambda i)$  to filter high-frequency stochastic noise

## OVERLAPPING RECEDING-HORIZON TRAJECTORY CHUNKS (H = 8 STEPS)



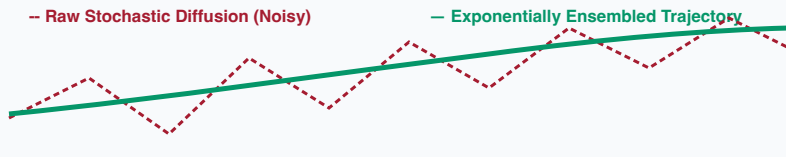
## EXPONENTIAL ENSEMBLING

$$\hat{a}_t = ( \sum w_i a_t^{(i)} ) / ( \sum w_i )$$

$$w_i = \exp( -\lambda \cdot (t - t_0^{(i)}) )$$

- **Decay Rate:**  $\lambda = 0.15$
- **Recency:** Weights newest token
- **Filtering:** Suppresses jitter

## TEMPORAL FILTERING: RAW STOCHASTIC NOISE VS. ENSEMBLE



- Temporal ensembling acts as a zero-phase low-pass filter, eliminating policy jitter.

## ✓ PHYSICAL SYSTEMS ADVANTAGES OF ENSEMBLING

1. **Smooth Inter-Chunk Handoff:** Prevents step jumps across chunk horizons.
2. **Inertial Compliance:** Blends future intention with immediate momentum.
3. **MCU Feasibility:** Enables bare-metal spline interpolator to compute  $C^2$  derivatives.
4. **Reduced Gearbox Stress:** Decreases peak motor current ripple by up to **68%**.